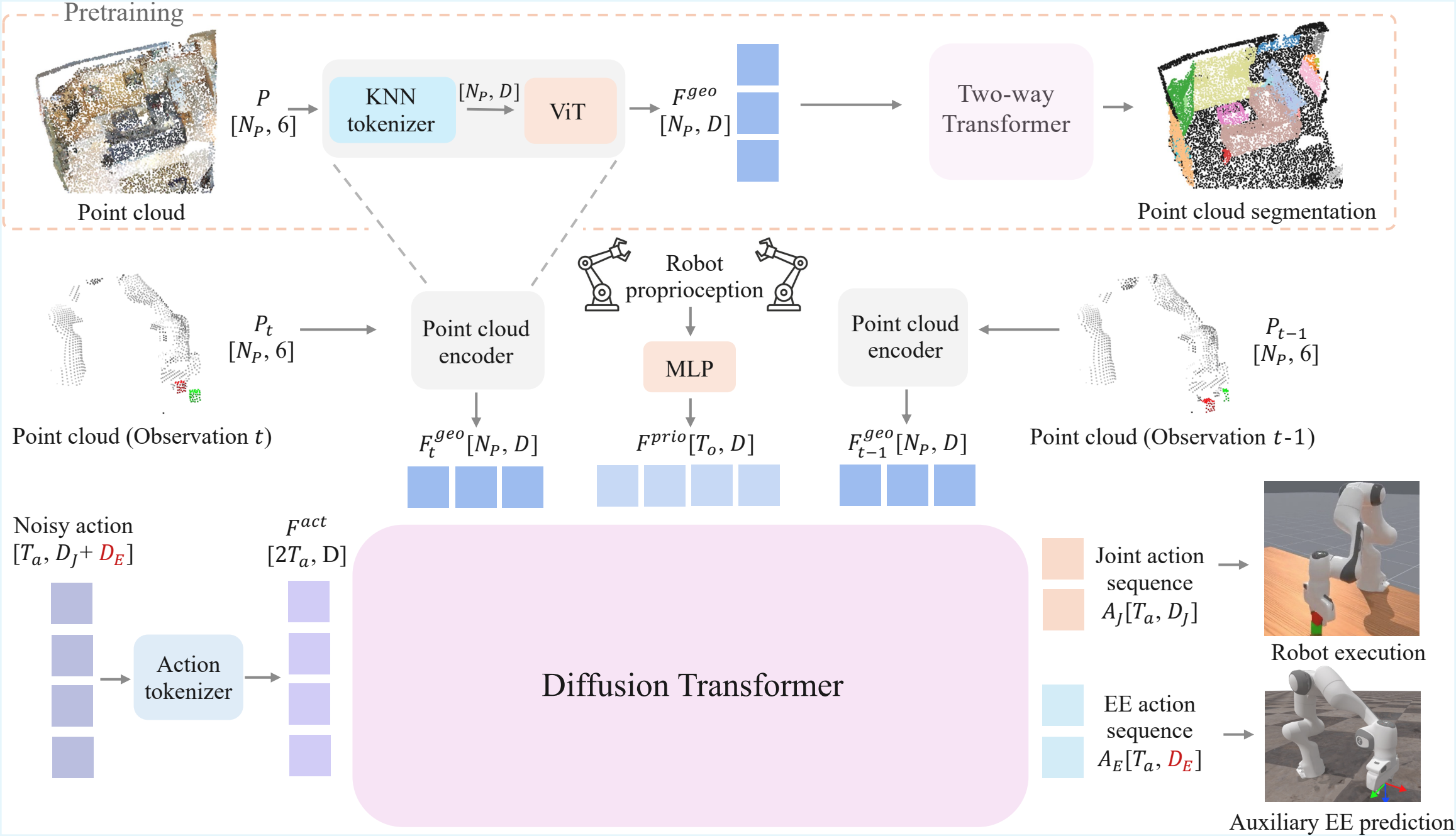
Pretraining
Point cloud
P
$[N_P, 6]$
KNN tokenizer
$[N_P, D]$
ViT
F^{geo}
$[N_P, D]$
Two-way Transformer
Point cloud segmentation
Robot proprioception
P_t
$[N_P, 6]$
Point cloud encoder
MLP
Point cloud encoder
P_{t-1}
$[N_P, 6]$
Point cloud (Observation t)
$F_t^{geo}[N_P, D]$
$F^{prio}[T_o, D]$
$F_{t-1}^{geo}[N_P, D]$
Point cloud (Observation t-1)
Noisy action
$[T_a, D_J + D_E]$
F^{act}
$[2T_a, \mathrm{D}]$
Action tokenizer
Diffusion Transformer
Joint action sequence
$A_J[T_a, D_J]$
Robot execution
EE action sequence
$A_E[T_a, D_E]$
Auxiliary EE prediction